Topic-wise Practice Questions

1. Introduction & Basic Programs (Easy)

1. Write a program to print "Hello World".
2. Print your name, age, and city.
3. Swap two numbers using a third variable.
4. Swap two numbers without using a third variable.
5. Find the sum of two numbers.
6. Find the product of three numbers.
7. Calculate the area of a rectangle.
8. Calculate the area of a circle.

2. Variables & Data Types

1. Input two integers and print their sum.
2. Find the average of three numbers.
3. Convert Celsius to Fahrenheit.
4. Find the square and cube of a number.
5. Calculate simple interest.
6. Convert seconds into hours, minutes, and seconds.

3. Operators

1. Check whether a number is even or odd.
2. Find the largest of two numbers.
3. Find the smallest of two numbers.
4. Check whether a number is positive, negative, or zero.
5. Find the maximum of three numbers.
6. Check whether a year is a leap year.
7. Check whether a character is a vowel or consonant.
8. Check whether a number is divisible by both 5 and 11.

4. if-else Statements

1. Check whether a student is pass or fail.
2. Find the grade based on marks.
3. Check whether a person is eligible for voting.
4. Find the largest among three numbers.
5. Check whether a character is alphabet, digit, or special character.
6. Check whether a triangle is valid.
7. Check whether a character is an alphabet.

5. Switch Case

1. Create a calculator using switch.
2. Print weekday using number.
3. Print month name using month number.
4. Check vowel using switch.
5. Find area of different shapes using switch.
6. Menu-driven arithmetic program.
7. ATM menu simulation.

6. Loops

1. Print numbers from 1 to 100.
2. Print even numbers from 1 to 100.
3. Find the sum of first N natural numbers.
4. Find factorial of a number.
5. Reverse a number.
6. Check palindrome number.
7. Count digits in a number.
8. Find the sum of digits.

7. Pattern Questions

```
*
**
***
****
*****
*****
****
***
**
*

1
12
123
1234
12345

A
AB
ABC
ABCD
ABCDE

    *
   ***
  *****
 *******
*********
```

More Patterns

- Hollow Square
- Hollow Rectangle
- Floyd Triangle
- Pascal Triangle
- Diamond Pattern

8. Functions

1. Function to find square.
2. Function to find cube.
3. Function to check even/odd.
4. Function to check prime.
5. Function to find factorial.
6. Function to reverse a number.
7. Function to find GCD.
8. Function to find LCM.
9. Function returning largest number.

9. Arrays (1D)

1. Input and print array.
2. Find sum of array.
3. Find average.
4. Find largest element.
5. Find smallest element.
6. Search an element.
7. Count even numbers.
8. Count odd numbers.
9. Reverse array.
10. Sort array in ascending order.
11. Sort array in descending order.
12. Find second largest element.
13. Remove duplicate elements.
14. Merge two arrays.

10. 2D Arrays

1. Input and print matrix.
2. Add two matrices.
3. Subtract two matrices.
4. Multiply two matrices.
5. Transpose a matrix.
6. Find row-wise sum.
7. Find column-wise sum.
8. Find diagonal sum.
9. Find largest element in matrix.

11. Strings

1. Find string length.
2. Reverse a string.
3. Check palindrome string.
4. Count vowels.
5. Count consonants.
6. Count words.
7. Convert uppercase to lowercase.
8. Convert lowercase to uppercase.
9. Compare two strings.
10. Concatenate two strings.
11. Remove spaces.
12. Find frequency of characters.

12. Pointers

1. Print value using pointer.
2. Swap using pointers.
3. Sum using pointers.
4. Reverse array using pointers.
5. Find largest element using pointers.
6. Pointer arithmetic examples.
7. Function using pointers.

13. Object-Oriented Programming (Moderate)

1. Create Student class.
2. Create Employee class.
3. Constructor example.
4. Destructor example.
5. Function overloading.
6. Operator overloading.
7. Single inheritance.
8. Multilevel inheritance.
9. Multiple inheritance.
10. Hybrid inheritance.
11. Virtual function example.
12. Abstract class example.
13. Friend function.
14. Static member.

14. File Handling

1. Write data into file.
2. Read data from file.
3. Copy file contents.
4. Count characters.
5. Count words.
6. Count lines.
7. Append data.
8. Search word in file.
9. Store student records.
10. Read student records.

Topic 1: Class & Object (Basic)

1. Create a `Student` class with data members:
    - Roll Number
    - Name
    - Marks

    Create an object and display the details.

2. Create an `Employee` class with employee ID, name, and salary.
3. Create a `Mobile` class with brand, RAM, and storage.
4. Create a `BankAccount` class with account number and balance. Display account details.
5. Create a `Rectangle` class and calculate area and perimeter.
6. Create a `Product` class and display product details.
7. Create a `Laptop` class with specifications.

Topic 2: Member Functions

1. Create a class to perform addition of two numbers.
2. Create a class to find factorial of a number.
3. Create a class to check whether a number is prime.
4. Create a class to reverse a number.
5. Create a class to check palindrome.
6. Create a class that calculates simple interest.
7. Create a class that accepts marks and calculates grade.
8. Create a class that finds largest among three numbers.

9. Create a class that performs calculator operations.

Topic 3: Constructor

1. Create a parameterized constructor.
2. Create both default and parameterized constructors.
3. Create multiple constructors using constructor overloading.
4. Display a message using constructor.
5. Initialize student details using constructor.
6. Create a constructor for bank account.
7. Calculate rectangle area using constructor.

Topic 4: Destructor

1. Create a class with constructor and destructor.
2. Display a message when object is destroyed.
3. Create multiple objects and observe destructor calls.
4. Create a dynamic object and delete it.
5. Demonstrate constructor-destructor execution order.

Topic 5: Encapsulation

1. Create a class with private data members and public setter/getter methods.
2. Create a Bank Account class where balance is private.
3. Create Employee class with private salary.
4. Create Student class using encapsulation.
5. Validate age using setter method.

Topic 6: Inheritance (Single)

1. Create Person → Student.
2. Create Animal → Dog.
3. Create Employee → Manager.
4. Display inherited data members.
5. Add additional data members in derived class.
6. Call base class function from derived class.
7. Constructor in inheritance.
8. Access inherited members.

Topic 7: Multilevel Inheritance

1. Person → Employee → Manager
2. Animal → Mammal → Dog
3. Student → Graduate → PostGraduate

Topic 8: Multiple Inheritance

1. Teacher + Researcher → Professor
2. Father + Mother → Child
3. Sports + Academics → Student

Topic 9: Hierarchical Inheritance

1. Shape → Circle, Rectangle, Triangle
2. Animal → Dog, Cat, Cow

3. Employee → Developer, Tester, Manager

Topic 10: Function Overloading

1. Add two integers.
2. Add three integers.
3. Add two float values.
4. Find area of square.
5. Find area of rectangle.

Compile Time Polymorphism

1. Function overloading.
2. Constructor overloading.
3. Operator overloading.
4. Calculator using overloading.
5. Area calculation using overloaded functions.

Run Time Polymorphism

6. Base class Shape and derived classes Circle and Rectangle using virtual function.
7. Employee salary calculation using virtual function.
8. Animal sound using virtual function.
9. Banking system using virtual function.
10. Vehicle information using virtual function.

Topic 13: Friend Function

1. Find sum of two private numbers.
2. Display private data.
3. Swap private values.
4. Compare two objects.
5. Access multiple classes using friend function.

Topic 14: Static Members

1. Count total objects created.
2. Static data member example.
3. Static member function example.
4. Bank account object counter.
5. Student object counter.

Topic 15: Abstract Class

1. Shape abstract class.
2. Vehicle abstract class.
3. Animal abstract class.

Topic 16: Mini OOP Projects (Moderate)

1. Student Management System
2. Library Management System
3. Bank Management System
4. Employee Payroll System
5. Hotel Booking System